



RT600

Dual-core microcontroller with 32-bit Cortex®-M33 and Xtensa HiFi4 Audio DSP CPUs; Up to 4.5 MB SRAM; FlexSPI with cache and dynamic decryption; High-speed USB device/host + Phy; 12-bit 1 Msamples/s ADC; Analog Comparator; Audio subsystems supporting up to 8 DMIC channels; SDIO/eMMC; AES/SHA/Crypto M33 coprocessor; PUF key generation

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Product data sheet

1. General description

The RT600 is a family of dual-core microcontrollers for embedded applications featuring an Arm Cortex-M33 CPU combined with a Cadence Xtensa HiFi4 advanced Audio Digital Signal Processor CPU. The Cortex-M33 includes two hardware coprocessors providing enhanced performance for an array of complex algorithms. The family offers a rich set of peripherals and very low power consumption.

The Arm Cortex-M33 is a next generation core based on the ARMv8-M architecture that offers system enhancements, such as ARM TrustZone® security, single-cycle digital signal processing, and a tightly-coupled coprocessor interface, combined with low power consumption, enhanced debug features, and a high level of support block integration. The ARM Cortex-M33 CPU employs a 3-stage instruction pipe and includes an internal prefetch unit that supports speculative branching. A hardware floating-point processor is integrated into the core. On the RT600, the Cortex-M33 is augmented with two hardware coprocessors providing accelerated support for additional DSP algorithms and cryptography.

The Cadence Xtensa HiFi 4 Audio DSP engine is a highly optimized audio processor designed especially for efficient execution of audio and voice codecs and pre- and post-processing modules. It supports four 32x32-bit MACs, some support for 72-bit accumulators, limited ability to support eight 32x16-bit MACs, and the ability to issue two 64-bit loads per cycle. There is a floating point unit providing up to four single-precision IEEE floating point MACs per cycle.

The RT600 provides up to 4.5 MB of on-chip SRAM (plus an additional 128 KB of tightly-coupled HiFi4 ram) and several high-bandwidth interfaces to access off-chip flash. The FlexSPI flash interface supports two channels and includes a 32 KB cache and an on-the-fly decryption engine. The RT600 is designed to allow the Cortex-M33 to operate at frequencies of up to 300 MHz and the HiFi4 DSP to operate at frequencies of up to 600 MHz.

1.1 Peripherals

The peripheral complement includes an FlexSPI flash interface with two channels, two SDIO/eMMC interfaces, a high-speed USB device/host with on-chip PHY, a 12-bit, 1 MSamples/sec ADC with temperature sensor, an analog comparator, AES256 and Hash engines with Physical Unclonable Function (PUF) key generation, a digital microphone



interface supporting up to eight channels and Voice Activation Detect, one I3C interface, one high-speed SPI interface and seven configurable serial interfaces that can be configured as a USART, SPI, I2C or I2S bus interface, each including a FIFO. When configured as USARTs the serial interfaces have the option to operate in deep-sleep mode using the 32 kHz oscillator or an external clock. There is a dedicated fractional baud rate generator for each of the serial interfaces.

Timing peripherals include one advanced, 32-bit SCTimer/PWM module, five general purpose 32-bit timer/counters with PWM capability, a 24-bit, multiple-channel multi-rate timer, two windowed watchdog timers, a system tick timer with capture capability, and a Real-time clock module with independent power and a dedicated oscillator. A common OS Event Timer is provided for synchronized event generation and timestamping between the two CPUs.

There are two general purpose DMA engines which can service most of the peripherals described in this section. The two DMA engines may be assigned to different CPUs and/or one may be used for secure operations, the other for non-secure.

Mailboxes and hardware semaphores are provided to facilitate inter-core communication. A variety of oscillators and PLLs are available as clock sources throughout the system.

1.2 Shared system SRAM

The entire system SRAM space of up to 4.5 MB is divided into up to 30 separate partitions, which are accessible to both CPUs, both DMA engines, and all other AHB bus masters. The HiFi4 CPU accesses the RAM via a dedicated 256-bit interface. Cache (with single-cycle access) is provided on this interface to improve performance. All other masters, including the Cortex-M33 processor and the DMA engines, access RAM via the main 32-bit AHB bus. These accesses are all single-cycle. Hardware interface modules arbitrate access to each RAM partition between the HiFi4 and the AHB bus.

Under software control, each of the 30 individual SRAM partitions can be used exclusively as code or as data, dedicated either CPU, or shared among the various masters. Each partition can be independently placed in a low-power retention mode or powered off entirely.

In addition to the shared SRAM, a total of 128 KB (64 KB code, 64 KB data) of local, Tightly-Coupled Memory (TCM) is provided for the exclusive use of the HiFi4 DSP processor. Access to this memory is single-cycle.

2. Features and benefits

- Control processor core
 - ◆ Arm Cortex-M33 processor, running at frequencies of up to 300 MHz.
 - ◆ Arm TrustZone.
 - ◆ Arm Cortex-M33 built-in Memory Protection Unit (MPU) supporting eight regions
 - ◆ Hardware Floating Point Unit (FPU).
 - ◆ Arm Cortex-M33 built-in Nested Vectored Interrupt Controller (NVIC).
 - ◆ Non-maskable Interrupt (NMI) input.

14.8 I²C-bus**Table 40. Dynamic characteristic: I²C-bus pins^[1]** $T_{amb} = 0\text{ }^{\circ}\text{C to }+85\text{ }^{\circ}\text{C}; 1.71\text{ V to }3.6\text{ V.}^{[2]}$

Symbol	Parameter		Conditions	Min	Max	Unit
f _{SCL}	SCL clock frequency		Standard-mode	0	100	kHz
			Fast-mode	0	400	kHz
			Fast-mode Plus	0	1	MHz
t _{LOW}	LOW period of the SCL clock	[8]	Standard-mode	4.7	-	μs
			Fast-mode	1.3	-	μs
			Fast-mode Plus	0.5	-	μs
t _{HIGH}	HIGH period of the SCL clock	[8]	Standard-mode	4.0	-	μs
			Fast-mode	0.6	-	μs
			Fast-mode Plus	0.26	-	μs
t _{HD;DAT}	data hold time	[3][4][5]	Standard-mode	0	-	μs
			Fast-mode	0	-	μs
			Fast-mode Plus	0	-	μs
t _{SU;DAT}	data set-up time	[6][7]	Standard-mode	250	-	ns
			Fast-mode	100	-	ns
			Fast-mode Plus	50	-	ns

- [1] Guaranteed by design. Not tested in production.
- [2] Parameters are valid over operating temperature range unless otherwise specified. See the I²C-bus specification *UM10204* for details.
- [3] t_{HD;DAT} is the data hold time that is measured from the falling edge of SCL; applies to data in transmission and the acknowledge.
- [4] Ensure SCL drops below 0.3VDD on falling edge before SDA crosses into the indeterminate range of 0.3 VDD to 0.7 VDD.
NOTE: For controllers that cannot observe the SCL falling edge then independent measurement of the time for the SCL transition from static high (VDD) to 0.3 VDD should be used to insert a delay of the SDA transition with respect to SCL.
- [5] The maximum t_{HD;DAT} could be 3.45 μs and 0.9 μs for Standard-mode and Fast-mode but must be less than the maximum of t_{VD;DAT} or t_{VD;ACK} by a transition time. This maximum must only be met if the device does not stretch the LOW period (t_{LOW}) of the SCL signal. If the clock stretches the SCL, the data must be valid by the set-up time before it releases the clock.
- [6] t_{SU;DAT} is the data set-up time that is measured with respect to the rising edge of SCL; applies to data in transmission and the acknowledge.
- [7] A Fast-mode I²C-bus device can be used in a Standard-mode I²C-bus system but the requirement t_{SU;DAT} = 250 ns must then be met. This will automatically be the case if the device does not stretch the LOW period of the SCL signal. If such a device does stretch the LOW period of the SCL signal, it must output the next data bit to the SDA line t_{r(max)} + t_{SU;DAT} = 1000 + 250 = 1250 ns (according to the Standard-mode I²C-bus specification) before the SCL line is released. Also the acknowledge timing must meet this set-up time.
- [8] The MSTTIME register allows programming of certain times for the clock (SCL) high and low times. Please see RT600 user manual UM11147 for further details.
- [9] Fall-time spec can be found in [Table 30 "Static characteristics: pin characteristics"](#).

14.17 MIPI I³C interface

Unless otherwise specified, MIPI I³C specifications are timed to/from the V_{IH} and /or V_{IL} signal points.

Table 52. MIPI I³C specifications when communication with legacy I²C devices [\[1\]](#)

Symbol	Characteristic	400 kHz/Fast mode		1 MHz/ Fast+ mode		Unit
		Min	Max	Min	Max	
f _{SCL}	SCL Clock Frequency	0	0.4	0	1	MHz
t _{SU_STA}	Set-up time for a repeated START condition	600	-	260	-	ns
t _{HD_STA}	Hold time (repeated) START condition	600	-	260	-	ns
t _{LOW}	LOW period of the SCL clock	1300	-	500	-	ns
t _{HIGH}	HIGH period of the SCL clock	600	-	260	-	ns
t _{SU_DAT}	Data set-up time	100	-	50	-	ns
t _{HD_DAT}	Data hold time for I ² C bus devices	0	-	0	-	ns
t _f	Fall time of SDA and SCL signals	20*(V _{dd} /5.5 v)	300	20*(V _{dd} /5.5 v)	120	ns
t _r	Rise time of SDA and SCL signals	20	300	-	120	ns
t _{SU_STO}	Set-up time for STOP condition	600		260	-	ns
t _{BUF}	Bus free time between STOP and START condition	1.3		0.5	-	μs
t _{SPIKE}	Pulse width of spikes that must be suppressed by the input filter	0	50	0	50	ns

[1] Based on simulation, not tested in production.

Table 53. MIPI I³C open drain mode specifications [\[1\]](#)

Symbol	Characteristic	Min	Max	Unit	Notes	Symbol
t _{LOW_OD}	LOW period of the SCL clock	200	-	ns		t _{LOW_OD}
t _{HIGH}	HIGH period of the SCL clock (for Mixed Bus)	-	41	ns		t _{HIGH}
	HIGH period of the SCL clock (for Pure Bus)	24	-	ns		
t _{fDA_OD}	Fall time of SDA signal	-	12	ns		t _{fDA_OD}
t _{SU_OD}	Data set-up time during open drain mode	3	-	ns		t _{SU_OD}
t _{CAS}	Clock after START (S) Condition					t _{CAS}
	ENTAS0	38.4 nano	1 μ	seconds		
	ENTAS1		100 μ	seconds		
	ENTAS2		2 milli	seconds		
	ENTAS3		50 milli	seconds		
t _{CBP}	Clock before STOP (P) condition	t _{CAS} (min)/2	-	seconds		t _{CBP}
t _{MMOverlap}	Current master to secondary master overlap time during hand off	t _{DIG_OD_L}	-	ns		t _{MMOverlap}
t _{AVAL}	Bus available condition	1	-	μs		t _{AVAL}
t _{IDLE}	Bus idle condition	200	-	μs		t _{IDLE}
t _{MMLock}	Time interval where new master not driving SDA low	t _{AVAL}	-	μs		t _{MMLock}

[1] Based on simulation, not tested in production.

Table 54. MIPI I3C push-pull specifications for SDR and HDR-DDR modes [1](#)

Symbol	Characteristic	Min	Typ	Max	Unit	Notes
f_{SCL}	SCL Clock Frequency	0.01	12.5	13	MHz	
t_{LOW}	LOW period of the SCL clock	24	-	-	ns	
t_{DIG_L}		32	-	-	ns	
t_{HIGH_MIXED}	High period of the SCL clock for a mixed bus	24	-	-	ns	
$t_{DIG_H_MIXED}$		32	-	45	ns	1
t_{HIGH}	HIGH period of the SCL clock	24	-	-	ns	
t_{DIG_H}		32	-	-	ns	
t_{SCO}	Clock in to data out for a slave	-	-	12	ns	
t_{CR}	SCL clock rise time	-	-	$150e06 * 1 / f_{SCL}$ (capped at 60)	ns	
t_{CF}	SCL clock fall time	-	-	$150e06 * 1 / f_{SCL}$ (capped at 60)	ns	
t_{HD_PP}	SDA signal data hold					
	Master mode	$t_{CR} + 3$ and $t_{CF} + 3$	-	-	ns	
	Slave mode	0	-	-	ns	
t_{SU_PP}	SDA signal setup	3	-	-	ns	
t_{CASr}	Clock after repeated START (Sr)	$t_{CAS(min)}/2$	-	-	ns	
t_{CBSr}	Clock before repeated START (Sr)	$t_{CAS(min)}/2$	-	-	ns	
C_b	Capacitive load per bus line	-	-	50	pF	

[1] Based on simulation, not tested in production.

[2] When communication with an I3C Device on a mixed Bus, the t_{DIG_H} period must be constrained in order to make sure that I2C devices do not interpret I3C signaling as valid I2C signaling.

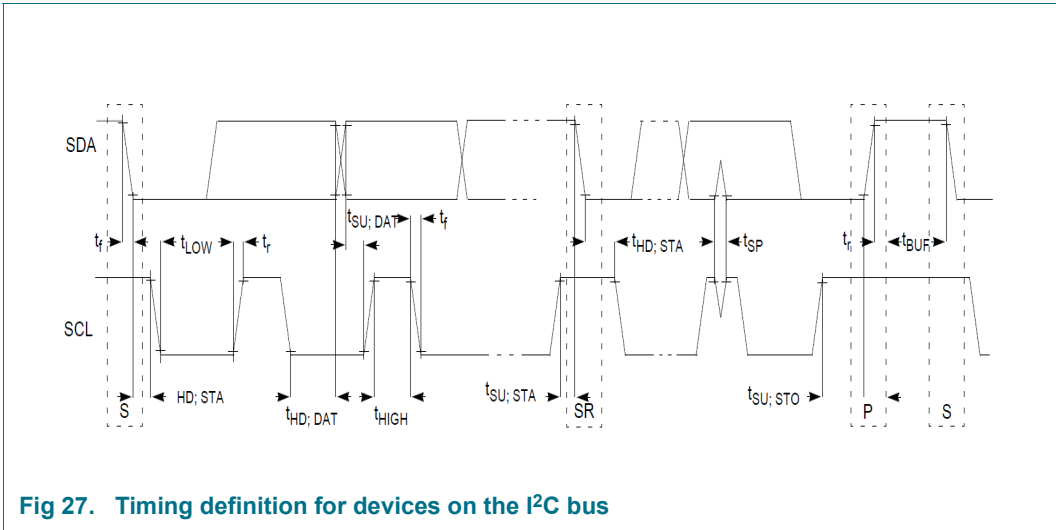


Fig 27. Timing definition for devices on the I2C bus